

Lab 001 Notes — Windows Event Investigation

Lab Objective

The objective of this lab was to learn how Windows records system and security activity and how a SOC analyst can use Event Viewer to examine that activity.

Environment

Operating system: Windows 11 Home

Computer type: 64-bit desktop computer

Memory: 16 GB RAM

Processor: Intel Core i7-9700

Tools used: Windows Event Viewer and Windows Snipping Tool

Concepts Learned

An event is a record of something that occurred on a computer.

A log is a collection of related events.

Windows Event Viewer is a built-in Windows tool that allows users and analysts to examine those logs.

The System log contains events related to Windows components, services, drivers, startup, shutdown, and system operation.

The Security log contains security-related records such as successful logons, failed logons, account activity, and permission-related events.

Event Metadata

The Event Level describes how Windows categorized the event, such as Information, Warning, Error, or Critical.

The Source identifies the program or Windows component that generated the event.

The Event ID identifies the specific type of recorded activity.

The Date and Time show when the event was recorded.

Event 10016 Observation

I observed a Windows System event with the following information:

Level: Warning

Source: DistributedCOM

Event ID: 10016

This event taught me that a warning is not automatically evidence of malware or a security compromise. It indicates that Windows recorded something that may deserve attention, but the event must be interpreted in context.

An Error also does not automatically mean that a computer was attacked. It may represent a program, driver, update, or other component that failed to work correctly.

Event 4624 Observation

I searched the Windows Security log for Event ID 4624.

The event stated:

An account was successfully logged on.

The Subject information included:

Security ID: SYSTEM

Account Name: A computer account ending with a dollar sign

Account Domain: WORKGROUP

The SYSTEM account represents Windows performing internal operating system activity.

The dollar sign at the end of the account name was an important clue that the event involved a computer account rather than a normal human username.

Analyst Interpretation

Event ID 4624 confirms that a successful logon occurred, but it does not prove that a human user signed in.

Successful logon events can be created by:

- Human users
- Windows services
- Applications
- Scheduled processes
- The Windows operating system
- Computer accounts

Additional evidence and surrounding context are required before deciding whether a successful logon is suspicious or malicious.

Main Lesson

A SOC analyst should not declare an incident based on one Event ID or one warning.

The analyst must examine:

- The account involved
- The time of the activity
- The logon type
- The source computer
- Related events
- Processes that ran
- Network activity
- Whether the behavior was expected

Cybersecurity investigations should be based on evidence, patterns, and context rather than assumptions.

Guided-Lab Disclosure

I completed this beginner lab through guided instruction. I personally performed the navigation, searched the logs, observed the evidence, answered the investigation questions, and documented my understanding.